

Timo Menzel

GRAPHICS ENGINEERING • DIGITAL HUMAN SYSTEMS

PhD candidate and research assistant at TU Dortmund University, specializing in 3D avatar scanning and personalization using consumer hardware. I develop end-to-end pipelines transforming smartphone captures into realistic full-body avatars and personalized facial animations for real-time and VR applications. My work focuses on graphics engineering and 3D reconstruction for digital humans.

CONTACT



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SKILLS

Graphics & Vision

Geometry Processing Photogrammetry
Non-rigid Registration Kinematics
Blendshapes Mesh Reconstruction

Programming

C++ Swift Python C#

Frameworks

ARKit OpenCV Eigen
RealityKit MediaPipe OpenGL
iOS macOS Linux
GoogleTest XCTest

Tools

Git Unity SQL CMake GCC

Systems

3D Pipelines Client-Server Systems

Languages

German (Native)
English (Fluent)
French (Rudimentary)

EXPERIENCE

Graphics & Geometry Group

Bielefeld University → TU Dortmund University

RESEARCH ASSISTANT / PHD CANDIDATE

May 2020 - Present

- Work continuously with the same research group, which relocated from Bielefeld University to TU Dortmund University in 2020.
- Conduct research in 3D avatar scanning and personalization from smartphone data
 - Develop automated pipelines for avatar reconstruction projects.
 - Collaborate with interdisciplinary teams to evaluate new methods and system usability.
 - Publish findings from current work in peer-reviewed venues.

SELECTED PROJECTS

AVATARS FOR THE MASSES

2023 - 2025

- Designed and implemented a fully automated end-to-end pipeline for VR-ready full-body avatar reconstruction from smartphone captures.
- Built a mobile capture app and scalable server pipeline enabling non-experts to acquire high-quality scans in uncontrolled environments.
- Integrated guided capture, segmentation, landmark detection, photogrammetry, and template fitting into a workflow producing high-quality avatars in 20 minutes.
- Evaluations showed avatars comparable to expert scans, demonstrating accessibility and feasibility for large-scale use.
- **Publication:** "Avatars for the Masses: Smartphone-Based Reconstruction of Humans for Virtual Reality" – <https://doi.org/10.3389/frvir.2025.1583474>
- **Project Page:** <https://avatars.cs.tu-dortmund.de>

COMPENSATING MOTION-INDUCED ERRORS

2024 - 2025

- Extended the "Avatars for the Masses" pipeline to reduce motion artifacts in sequential smartphone captures, improving reconstruction quality.
- Developed a sub-scan processing strategy dividing image sequences into temporal segments and integrating silhouette constraints.
- Improved accuracy in critical regions (e.g., arms) while maintaining a simple, user-friendly capture workflow.
- **Publication:** "Compensating Motion-Induced Errors in Smartphone-Based VR Avatar Reconstruction" – <https://doi.org/10.1145/3756884.3765995>

AUTOMATED BLENDSHAPE PERSONALIZATION

2021 - 2022

- Developed an automated pipeline for personalized facial blendshapes from smartphone captures using ARKit.
- Generated personalized blendshape masks via position-based optimization and integrated them into realistic avatars.
- Achieved significantly more accurate reconstruction of facial expressions compared to prior methods.
- **Publication:** "Automated Blendshape Personalization for Faithful Face Animations Using Commodity Smartphones" – <https://doi.org/10.1145/3562939.3565622>

● EDUCATION

○ TU Dortmund University

PHD IN COMPUTER SCIENCE

Dortmund, Germany

2020 - Expected Summer 2026

- Research focus on 3D avatar reconstruction and personalization from smartphone data.
- Published multiple first- and shared first-author papers in ACM VRST and Frontiers in VR.
- Developed automated pipelines for avatar reconstruction and blendshape personalization for non-expert users.

○ Bielefeld University

M.Sc. IN INTELLIGENT SYSTEMS

Bielefeld, Germany

2018-2020

- Interdisciplinary program covering AI, intelligent systems, and human-technology interaction.
- Focus on developing and deploying adaptive, interactive systems using modern software engineering and scientific methods.

○ Bielefeld University

B.Sc. IN INFORMATICS FOR THE NATURAL SCIENCES

Bielefeld, Germany

2015-2018

- Interdisciplinary computer science curriculum focused on applying computational methods to natural science problems (physics, chemistry, biology).
- Emphasis on scientific computing, data analysis, and model-driven software development for research and industry contexts.

● PUBLICATIONS

Friedemann Runte, Timo Menzel, Ulrich Schwanecke, and Mario Botsch. *Compensating Motion-Induced Errors in Smartphone-Based VR Avatar Reconstruction*. In: *Proceedings of the 31st ACM Symposium on Virtual Reality Software and Technology*. VRST '25. 2025.

DOI: [10.1145/3756884.3765995](https://doi.org/10.1145/3756884.3765995).

Timo Menzel, Erik Wolf, Stephan Wenninger, Niklas Spinczyk, Lena Holderrieth, Carolin Wienrich, Ulrich Schwanecke, Marc Erich Latoschik, and Mario Botsch. *Avatars for the Masses: Smartphone-Based Reconstruction of Humans for Virtual Reality*. In: *Frontiers in Virtual Reality* 6 (2025).

DOI: [10.3389/frvir.2025.1583474](https://doi.org/10.3389/frvir.2025.1583474).

Peter Kullmann, Theresa Schell, Timo Menzel, Mario Botsch, and Marc Erich Latoschik. *Coverage of Facial Expressions and Its Effects on Avatar Embodiment, Self-Identification, and Uncanniness*. In: *IEEE Transactions on Visualization and Computer Graphics* 31 (2025).

DOI: [10.1109/TVCG.2025.3549887](https://doi.org/10.1109/TVCG.2025.3549887).

Peter Kullmann, Timo Menzel, Mario Botsch, and Marc Erich Latoschik. *An Evaluation of Other-Avatar Facial Animation Methods for Social VR*. In: *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. CHI EA '23. 2023.

DOI: [10.1145/3544549.3585617](https://doi.org/10.1145/3544549.3585617).

Timo Menzel, Mario Botsch, and Marc Erich Latoschik. *Automated Blendshape Personalization for Faithful Face Animations Using Commodity Smartphones*. In: *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology*. VRST '22. 2022.

DOI: [10.1145/3562939.3565622](https://doi.org/10.1145/3562939.3565622).